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PROGRESSIVE AMMUNITION PRESS SYSTEM

Abstract

Progressive ammunition press system, components thereof, and associated methods. The press system may include a press control system integrated with the press. The press may include a case carriage including a case carrier supported by a bearing. Case retainer gates may be provided for retaining cases in the case carrier. A bullet feeding system can be arranged to permit bullet feeding and seating in a single station of the press. A primer feeder can be self-contained, gear-driven, and/or recirculating.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a divisional of U.S. patent application Ser. No. 17/655,777, filed Mar. 21, 2022, which claims priority to U.S. Provisional Application No. 63/163,520, filed Mar. 19, 2021, the entireties of which are hereby incorporated by reference.

FIELD

[0002] The present disclosure generally relates to firearm ammunition manufacturing equipment, and more particularly to a progressive ammunition press system and associated components and methods.

BACKGROUND

[0003] Persons manufacturing or reloading firearm ammunition often use various types of equipment. A variety of types of ammunition presses are available for such purposes. Progressive presses permit a user to complete multiple operations in successive stations of the press.

SUMMARY

[0004] In one aspect, a progressive ammunition press system comprises a frame configured to be supported by a support surface. A case carrier is supported by the frame. The case carrier includes multiple case holders each configured to hold an ammunition case. The case carrier is moveable with respect to the frame to move the case holders to a plurality of stations. A tool head is supported by the frame. The tool head comprises a plurality of press accessory mounts. The tool head is moveable with respect to the case carrier. A drive system is operatively connected to the tool head to move the tool head with respect to the case carrier. An electrical power input port is supported by and moveable with the tool head. An electrical power output port supported by and moveable with the tool head and operatively coupled to the electrical power input port to receive power therefrom.

[0005] In another aspect, a progressive ammunition press system comprises a frame configured to be supported by a support surface. A case carrier supported by the frame includes multiple case holders each configured to hold an ammunition case. The case carrier is moveable with respect to the frame to move the case holders to a plurality of stations. A tool head supported by the frame comprises a plurality of press accessory mounts. The tool head is moveable with respect to the case carrier. A drive system is operatively connected to the tool head to move the tool head with respect to the case carrier. A communications output port is supported by and moveable with the tool head. A communications input port is supported by and moveable with the tool head and operatively coupled to the communications output port to transmit communication signals to the communications output port.

[0006] In yet another aspect, a progressive ammunition press system comprises a frame configured to be supported by a support surface. A case carrier supported by the frame includes multiple case holders each configured to hold an ammunition case. The case carrier is moveable with respect to the frame to move the case holders to a plurality of stations. A tool head supported by the frame comprises a plurality of press accessory mounts. The tool head is moveable with respect to the case carrier. A drive system is operatively connected to the tool head to move the tool head with respect to the case carrier. A communications port is supported by the frame. A press controller is supported by the tool head. A tangible storage medium is supported by the tool head and stores press controller executable instructions to generate a command signal to control an operation of the

ammunition press system based on a sensor signal received from the communications port.
[0007] Other objects and features of the present disclosure will be in part apparent and in part pointed out herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a front perspective of a progressive ammunition press system of the present disclosure;
[0009] FIG. 2 is an enlarged fragmentary perspective of the ammunition press system similar to the view of FIG. 1;
[0010] FIG. 3 is an enlarged rear fragmentary perspective of the ammunition press system;
[0011] FIG. 4 is a top view of an upper tool head of the ammunition press system;
[0012] FIG. 5 is a perspective of a frame and drive components of the ammunition press system;
[0013] FIG. 6 is a section of the frame with the drive components;
[0014] FIG. 7 is a perspective of drive components;
[0015] FIG. 8 is a perspective of a journal shaft;
[0016] FIG. 9 is a perspective of another journal shaft;
[0017] FIG. 10 is a perspective of a case carriage of the ammunition press system;
[0018] FIG. 11 is a top view of the case carriage;
[0019] FIG. 12 is an exploded perspective of the case carriage;
[0020] FIG. 13 is a fragmentary exploded perspective of the case carriage, frame, and drive components;
[0021] FIG. 14 is a top view of the case carriage installed on the frame;
[0022] FIG. 15 is a fragmentary exploded perspective of a ball detent for engagement with the case carriage;
[0023] FIG. 16 is a top view of the ball detent installed on the frame;
[0024] FIG. 17 is a fragmentary exploded perspective of case retainers and the frame;
[0025] FIG. 18 is a section of a case retainer and the frame;
[0026] FIG. 19 is a side view of the frame and drive components associated with an indexing carriage for indexing the case carrier;
[0027] FIG. 20 is a fragmentary exploded perspective of the frame, indexing carriage, and drive components;
[0028] FIG. 21 is an exploded perspective of the frame and the indexing carriage;
[0029] FIG. 22 is an enlarged perspective of the indexing carriage;
[0030] FIG. 23 is an exploded perspective of the indexing carriage;
[0031] FIG. 24 is a perspective of another embodiment of an indexing carriage;
[0032] FIG. 25 is a fragmentary exploded perspective of the frame, drive components, and the upper tool head;
[0033] FIG. 26 is a section of a depth stop installed on the upper tool head;
[0034] FIG. 27 is a top perspective of the tool head;
[0035] FIG. 28 is a bottom perspective of the tool head;
[0036] FIG. 29 is an exploded perspective of the tool head;
[0037] FIG. 30 is a schematic of a control system of the ammunition press system;
[0038] FIG. 31 is a fragmentary section of the ammunition press system showing a primer collection feature;
[0039] FIG. 32 is an enlarged fragmentary perspective of a primer collection tube extending from the frame;
[0040] FIG. 33 is a fragmentary perspective of the upper tool head having a support die installed

thereon;

[0041] FIG. **34** is an exploded perspective of the support die;

[0042] FIG. **35** is a fragmentary side view of the ammunition press system showing a lower tool head and associated drive components;

[0043] FIG. **36** is an exploded perspective of the frame, the lower tool head, and drive components;

[0044] FIG. **37** is a fragmentary exploded perspective of the frame, case carriage, and primer feeder;

[0045] FIG. **38** is an exploded perspective of the primer feeder;

[0046] FIG. **39** is an enlarged fragmentary section of an upper end of the primer feeder;

[0047] FIG. **40** is an enlarged fragmentary section of a lower end of the primer feeder;

[0048] FIG. **41** is an enlarged fragmentary section of the primer feeder mounted on the frame;

[0049] FIG. **42** is a top view of the primer feeder installed on the frame;

[0050] FIG. **43** is a fragmentary perspective of an upper end of a powder dispenser;

[0051] FIG. **44** is an enlarged exploded perspective of a low primer sensor assembly;

[0052] FIG. **45** is a fragmentary perspective of a lower end of a cord from the low primer sensor assembly connected to a port in the upper tool head;

[0053] FIG. **46** is a perspective of a primer follower;

[0054] FIG. **47** is an enlarged fragmentary section of the frame and primer feeder with an automatic primer detector system;

[0055] FIG. **48** is a perspective of components of the automatic primer detector system;

[0056] FIG. **49** is a side elevation of a powder dispenser in a non-actuated configuration;

[0057] FIG. **50** is a side elevation of the powder dispenser in an actuated configuration;

[0058] FIG. **51** is an enlarged fragmentary section of the powder dispenser in the non-actuated configuration;

[0059] FIG. **52** is an enlarged fragmentary elevation of the powder dispenser and frame showing a powder dispenser guide rod and reset feature;

[0060] FIG. **53** is an enlarged fragmentary section of a bullet feeder on the upper tool head;

[0061] FIG. **54** is an enlarged fragmentary elevation of the bullet feeder showing drive components;

[0062] FIG. **55** is an enlarged fragmentary section of an alternative embodiment of a bullet feeder;

[0063] FIG. **56** is a view similar to FIG. **55** but showing a shuttle of the bullet feeder moved to position a bullet in registration with a bullet seating die;

[0064] FIG. **57** is a view similar to FIG. **56** but showing a case entering the bullet feeder;

[0065] FIG. **58** is a view similar to FIG. **57** but showing the bullet being seated in the case;

[0066] FIG. **59** is an exploded perspective of the press system showing covers for installation on the frame;

[0067] FIG. **60** is a fragmentary exploded perspective of the ammunition press system showing a bracket and associated bin for receiving assembled rounds of ammunition;

[0068] FIG. **61** is a fragmentary perspective showing the bracket connected to the frame and the bin on the bracket; and

[0069] FIG. **62** is a fragmentary perspective of the ammunition press system including an automatic actuation mechanism.

[0070] Corresponding reference characters indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

[0071] Referring to FIGS. **1-3**, a progressive press system of the present disclosure is generally indicated by the reference number **10**. As explained in further detail below, the press system is configured to manufacture (e.g., reload) firearm ammunition. The press system can be configured to receive bulk ammunition component materials, such as ammunition cases (sometimes referred to as shells), primers, ammunition powder, and bullets, and assemble the components into ammunition cartridges. The press system includes several improvements relative to conventional

systems.

[0072] The press system **10** includes an ammunition press **12** and a plurality of press accessories. Referring to FIGS. **2** and **3**, the press includes a frame **14**, a case carriage **16** (see FIGS. **10-14**), an upper tool head **18**, a lower tool head **20**, and a drive system **22**. The frame **14** supports components of the press system and defines a base for resting on a support surface such as a table or reloading bench. The case carriage **16** is configured to carry cases to multiple stations for working the cases and/or assembling the ammunition cartridges. Accessories are mountable on the tool heads **18**, **20** to be associated with respective stations to which the cases are indexed. The upper tool head **18** is configured to hold accessories to move up and down above the cases held by the case carriage. The lower tool head **20** is configured to hold accessories (e.g., swaging and priming pins) to move up and down below the cases. The drive system **22** (broadly, “press driver”) is configured to turn the cases through the series of stations and to cause movement of the tool heads **18**, **20**. The drive system **22** includes an actuation lever **24** including a handle by which a user can manually operate the press **12**. Alternatively, an automatic actuator **26** including a motor **28** can be used, as described further below with reference to FIG. **62**. After each downward stroke of the tool head **18**, the cases are indexed to the next stations.

[0073] The press system **10** includes press accessories such as dies and/or tools, and other types of accessories, such as feeders and dispensers. The press accessories are mountable on the press **12**, such as on the frame **14** and/or on the tool heads **18**, **20**. As shown in FIG. **1**, the illustrated accessories include a case feeder **30**, a primer feeder **32**, and a powder dispenser **34**. Other accessories, such as a bullet feeder **36** (e.g., see FIGS. **53-58**), other types of dies and tools, etc., can be used as well, without departing from the scope of the present disclosure.

[0074] In the illustrated embodiment, the press **12** includes ten stations **S1-S10**, which will be explained initially with respect to FIG. **4**. It will be appreciated that the stations can be defined by various components of the press system **10**, including the frame **14**, the upper tool head **18**, and the lower tool head **20**. The stations **S1-S10** are locations to which the carriage **16** indexes cases to be worked on by accessories associated with the respective stations. As shown in FIG. **4**, the upper tool head **18** includes accessory mounts **38** (e.g., die mounts) corresponding to nine of the ten stations. The accessory mounts **38** can be configured to receive standard $\frac{7}{8}$ -14 unc dies (e.g., “threaded die mount opening”). In the illustrated embodiment, each station except for the first station **S1** has an associated threaded die mount opening. The first station **S1** is designated for feeding cases to the press. The remainder of the stations **S2-S10** can be configured as desired by the user by mounting accessories on the tool heads **18**, **20** to set up the press **12** for performing various operations. The cases are ejected from the case carriage **16** between the tenth station **S10** and the first station **S1**.

[0075] For example, the press system **10** can be configured to execute a single pass processing operation. The ten stations **S1-S10** provide ample locations to work cases such that a single pass processing operation is possible. In single pass processing, all operations required to load ammunition are performed in a single rotation of the case carriage **16**, from the case feed station **S1** to ejection, with minimal case processing (e.g., sorting, case lube (optional)) prior to inserting the case into the press system. Single pass processing can include (e.g., at a minimum) the following: decapping (i.e., removal of a spent primer), sizing (e.g., forming a fired case back to original size), priming (i.e., installation of a new primer), powder drop (i.e., dispensing of a measured powder charge), bullet feeding (feeding a bullet to a mouth of a case), bullet seating (pushing the bullet to a seated depth in the case mouth). Single pass processing can also include case lube, primer pocket swaging, case trimming, neck expansion (can be combined with powder drop), powder drop, powder check, bullet crimp (can be combined with bullet seating). Other steps and variations can be used. A variety of press accessories, such as dies and/or tools can be mounted on the press to implement a variety of single-pass processes, as desired by the user.

[0076] It will be appreciated that the press system **10** can be configured for multi-pass processing,

without departing from the scope of the present disclosure. In multi-pass processing, the same general operations as single-pass processing can be performed, but are accomplished for each case in two or more rotations of the case carriage **16**. Cases can be run as a batch through the press for initial processing. Then the configuration of the press **12** is changed, and the cases are run through a second time. Multi-pass processing provides an increased number of available stations if the user desires to perform operations at more than ten stations. Multi-pass processing can also provide smoother operation during powder metering and bullet feeding/seating if performed in a second pass, after performing decapping and resizing operations in a first pass.

[0077] Referring to FIG. 5, the frame **14** can be made of cast iron or another suitable material. The frame **14** includes an upper face and a lower base for resting on a support surface. The frame serves as supporting structure for components of the press system **10**.

[0078] Referring to FIGS. 1 and 5-9, the drive system **22** includes the actuator **24**, and a drive train for moving the tool heads **18**, **20** and indexing the case carriage **16**. The drive train (broadly, linkage) includes a left (first) journal shaft **40**, a right (second) journal shaft **42**, and a throw assembly **44**. The throw assembly **44** includes a crank assembly and a ram **46**. The ram **46** extends upward through a linear bearing **48** in the top of the frame and includes a threaded opening and flats (collectively, tool head mount) for mounting the upper tool head on the ram. The crank assembly includes left and right crank throws **50** connected to the ram via a link **52**. The link **52** is connected to the bottom of the ram **46** by a pivot connection and to the crank throws **50** by another pivot connection. The crank throws **50** are keyed with the respective journal shafts **40**, **42** for conjoint rotation with the journal shafts. The journal shafts **40**, **42** are received in bearings **54** in respective left and right sides of the frame **14**. The journal shafts **40**, **42** include lips to capture the bearings **54** in the frame **14** and center the throw assembly **44** between the bearings. Two screws **56** are threaded in the frame **14** to define stops to limit rotation of the crank throws **50** to about 180 degrees. The lower screw **56** defines the bottom travel extent of the ram **46**, and the top screw **56** defines the top travel extent of the ram. As shown in FIG. 1, the lever **24** is connected to a hub of the right journal shaft **42**. The arrangement is such that the actuator lever **24** can be moved in an actuation cycle including a downward stroke and an upward stroke. The journal shafts **40**, **42** and crank throws **50** pivot conjointly with the lever **24**. In the downward (actuation) stroke of the lever **24**, the ram **46** and thus the upper tool head **18** move downward, and in the upward (return) stroke, the ram and upper tool head move upward. The tool head **18** is lowered by the throw assembly **44** to reach bottom dead center at the rotational limit of the actuator lever defined by the lower screw **56**. The journal shafts **40**, **42** include cams **40A**, **42A** configured to transmit force from the journal shafts to other components of the drive system **22** to coordinate indexing of the case carriage **16** and movement of the lower tool head **20** and the upper tool head **18**. As the upper tool head **18** reaches bottom dead center, the lower tool head **20** reaches top dead center. After each downward movement of the upper tool head **18**, the drive system **22** indexes the case carriage **16** to move the cases to the next stations.

[0079] The drive system **22** can have other configurations without departing from the scope of the present disclosure. For example, components thereof (e.g., the linear bearing) may be omitted.

[0080] Referring to FIGS. 10-14, the case carriage **16** is configured to hold ten cases at a time for moving the cases through the ten stations S1-S10. The case carriage **16** includes an annular case carrier **58** (e.g., "shell plate"), a bearing **60**, and a gear **62**. The case carrier **58** includes a plurality of case holders **64** each configured to hold a single case. The case holders **64** each include a side opening permitting a case to slide laterally into the case holder, and a lip to capture a flange of the case to retain the case in the case holder. The cases can be loaded into the case carrier **58** by hand or using the case feeder. Different case carriers (with different sizes of case holders) are used for cases of different sizes. A kit of case carriers **58** can be used (interchangeable with the same bearing and gear). Alternatively, a kit of case carriages **16** each including a case carrier, bearing, and gear, can be used.

[0081] The bearing **60** includes an inner race **60A**, an outer race **60B**, and a plurality of balls captured therebetween. The bearing **60** facilitates rotation of the case carrier **58** to move the case holders **64** to the stations **S1-S10**.

[0082] The gear **62** is annular and includes a plurality of teeth **62A**. The gear is sized to receive the bearing in the gear. The teeth **62A** are configured to coordinate operation of the primer feeder with the case carriage, as explained in further detail below.

[0083] The case carrier **58** is connected to the bearing **60** and gear **62** using a plurality of fasteners. Heads of the fasteners abut the case carrier **58**, and opposite ends of the fasteners receive washers and nuts. As shown in FIG. **41**, the washers abut a bottom side of the gear **62** and a bottom side of the outer race **60B** of the bearing **60**. The nuts are tightened to press the upper side of the gear **62** and the upper side of the outer race **60B** against the bottom of the case carrier **58**. The nuts serve as indexing lugs **68** by which the drive system **22** rotates the case carrier **58**, as will be explained further below. Bottom ends of the nuts are rounded to facilitate engagement by the drive system **22**.

[0084] As shown in FIG. **13**, the upper face of the frame **14** defines an annular recess (broadly, “case carriage receiver”) configured to receive the case carriage **16**. The annular recess is defined in part by a central boss over which the case carriage is receivable. When the case carriage **16** is in the annular recess, the inner race **60A** of the bearing **60** rests on the frame **14** to locate the case carriage with respect to the frame. The case carriage **16** is secured to the frame by a case carriage retainer **70** (e.g., “bearing clamp”) secured to the frame via a plurality of fasteners. The case carriage retainer **70** engages the inner race **60A** of the bearing. The arrangement is such that the outer race **60B** of the bearing is free from engagement with the frame **14** to isolate the case carrier **58**, gear **62**, and indexing lugs **68** from the frame to facilitate their rotation with respect to the frame. The case carriage retainer **70** includes an arm **70A** configured to deflect cases out of the case carrier **58** between the tenth station **S10** and first station **S1** so a new case can be inserted in the case carrier at the first station.

[0085] A detent ball **72** and associated spring are captured in a pocket in the frame by a ball retainer **74** and are configured to hold the case carrier **58** in position with the case holders **64** at the stations **S1-S10**. The spring biases the ball **72** upward for reception in notches **58A** (broadly, “indexing openings”) spaced around a periphery of the case carrier **58** to index movement of the case carrier and retain the case holders **64** at the stations. When sufficient rotational force is applied to the case carrier **58** by the drive system **22**, the ball **72** is dislodged from one of the notches **58A** and rides on an underside of the case carrier until reaching the next notch.

[0086] Referring to FIGS. **14**, **17**, and **18**, case retainers **80** are provided at each of the stations except the case feeding station **S1** and the priming station **S5**. The case retainers **80** serve as gates to selectively permit removal/insertion of a case from/to the case carrier **58**. A user may want to remove or insert a case for inspection or to clear a jam. The case retainers **80** each comprise a spring-biased lug **80A** arranged to obstruct removal of a case from the case carrier **58** when the case is at one of the stations **S2-S4**, **S6-10**. Between the stations, the frame includes fences **84** (FIG. **14**) that obstruct removal the cases from exiting the case holders as the case carrier indexes.

[0087] As shown in FIGS. **17** and **18**, each case retainer **80** includes a screw **80B**, a spring **80C**, and a nut. The nuts (e.g., threaded “buttons”) form the lugs **80A** that block the case holders **64**. The screws **80B** have shafts extending upward through openings in the frame **14**. The springs **80C** are captured on the screw shafts between the nuts and the frame. The heads of the screws limit upward movement of the screws. In the illustrated embodiment, the springs **80C** are coil springs, and more particularly compression coil springs. However, other types of springs, such as tension springs can be used.

[0088] To move a lug **80A** (i.e., open the gate), the user presses downward on the lug with their hand or with the primer end of the case to resiliently compress the spring **80C**. This moves the retaining lug **80A** down into the upper face of the frame **14** to permit lateral passage of the case into or out of the case holder **64**. When pressure on the lug **80A** is released, the lug extends back to its at

rest or closed position. To remove a case, the lug **80A** is pressed downward and the case is extracted from the case carrier over the lug. The case is reinstalled by pressing down on the lug with the case, and then sliding the case on the surface of the frame into the case holder **64**. The case retainers **80** are vertically displaceable by applying a downward force to the retainers to deform the springs **80C**. The retainers **80** are spring loaded to be depressed vertically. It will be appreciated that the lugs **80A** permit one-handed installation of a case into a case holder **64** without requiring the user to touch the lugs by hand because the lug can be moved with the case held by the user. The case retainers **80** do not need to be separated from the frame **16** to permit a case to be inserted or removed. The case retainers **80** operate independently from each other. The case retainers **80** do not apply a biasing force to a case when the case is in a case holder **64**. When a case is in a case holder **64**, lateral force applied by a case to a case retainer **80** is not capable of compressing the spring **80C** or opening the case retainer. Other configurations can be used without departing from the scope of the present disclosure.

[0089] Referring to FIGS. **19-24**, the drive system **22** includes an index driver **84** configured to rotate the case carrier **58**. The index driver **84** includes an index carriage **86** driven by an index armature assembly **88** (broadly, “linkage”) installed over the cam **40A** of the left journal shaft **40**. The index carriage **86** moves along a track **90** defined by two guide rods. The index carriage **86** carries a bearing **92** (broadly, “connector”) received in a slot in the upper end of the index armature assembly **88**. The index armature assembly includes two bearings **94** (broadly, “wheels”) that ride on the cam **40A**. The cam profile is designed so that both bearings **94** are in contact with the cam **40A** through the entire 360 degree rotation of the left journal shaft **40**. The armature **88** remains stationary throughout most of the travel of the journal shaft **40** as the bearings **94** are in contact with opposing rounded portions of the cam **40A**. When the bearings **94** contact the flat portions of the cam profile, the armature **88** moves quickly to opposite extreme ends of its travel range. The connection of the index armature **88** to the index carriage **86** causes the index carriage to intermittently move horizontally along the track **90** when the index armature moves.

[0090] The index carriage **86** includes a catch **96** biased toward a case carrier engagement position by a torsion spring **98**. The catch **96** is configured to engage one of the indexing lugs **68** of the case carrier **16** to turn the case carrier **58**. The catch **96** has a finger arranged to pull an indexing lug to cause the case carrier to turn. When the index armature **88** moves the indexing carriage **86** to engage the next indexing lug **68**, the catch **96** disengages the previous indexing lug, and a ramp **96A** on the catch causes the catch to resiliently deflect over the next indexing lug. The index armature **88** then moves the catch to pull that indexing lug **68** to turn the case carrier again. The operation of the indexing carriage **86** is coordinated with movement of the tool heads **18, 20** via the drive system **22** so the case carrier **58** moves cases to the next stations when the accessories on the upper tool head are spaced upward from the cases and the accessories on the lower tool head are spaced downward. Each cycle of the index carriage **86** causes the case carrier **58** to rotate counter-clockwise about 36 degrees, which is the angular distance between adjacent case stations (e.g., **S2** and **S3**), between adjacent case holders **64**, and between adjacent notches **58A**.

[0091] As shown in FIG. **23**, the indexing carriage **86** includes a carriage body **86A** that receives two linear bearings **86B** held in the body by opposite screws and associated washers. The guide rods forming the track **90** extend through these linear bearings **86B**. The armature bearing **92** is secured to the body **86A** by a fastener. An upper fastener secures the catch **96** to the body **86A**, and a lower fastener threaded to the catch serves as a stop **86C** for the catch to oppose the bias of the torsion spring **98**. The torsion spring **98** is received over a hub **86D** to which the upper fastener is threaded.

[0092] In an alternative embodiment of the indexing carriage **86**, shown in FIG. **24**, the armature bearing **92** (connector) is horizontally adjustable with respect to the carriage body **86A**. This permits adjustment to locate the catch **96** to properly index movement of the case carrier **58**. In this embodiment, the carriage body **86A** includes an opening in which a slide **86E** is received. The

armature bearing **92** is mounted on the slide **86E**. When one or both fasteners are loose, the slide **86E** is movable in the carriage body **86A**, and the slide can be secured in an adjusted position by tightening the fasteners by threading them into the slide.

[0093] As shown in FIG. **25**, the upper tool head **18** is receivable on the mount of the ram **46** and secured to the ram via a fastener. A tool head guide **100** extends upward from the frame **14** to guide up and down movement of the tool head. In the illustrated embodiment, the tool head guide **100** comprises a rod secured to the frame. The guide **100** limits rotation of the tool head **18** about the ram **46** and forces the tool head die mounts **38** to be in vertical registration with the case stations. A coil spring **102** is provided on the guide **100** to facilitate upward return of the tool head **18**. The tool head **18** includes a guide receiver through which the guide **100** extends. As shown in FIGS. **25** and **27**, the guide receiver can comprise a bearing **103** in a pocket in the tool head.

[0094] In an alternative embodiment, shown in FIG. **26**, the guide receiver can comprise a tool head depth stop **104** (broadly, “torque brace”) to limit tool head rock or flex. It will be appreciated that the tool head **18** can be subjected to substantial force, such as in a resizing operation. The force on the tool head **18** at a station where resizing is happening may cause the tool head to flex or rock. Torque is induced in the tool head which can cause the tool head to rock relative to the shell plate as the case is resized. When the press system **10** is set up to perform several operations at different stations at the same time, varying displacements of the tool head **18** can affect operations offset from the resizing operation, such as bullet seating, and can prevent the cases from being completely resized. The tool head depth stop **104** is located on the tool head **18** generally on an opposing side of the ram **46** with respect to the second station **S2**, the recommended resizing station, and the fourth station **S4**, the recommended primer pocket swaging station. As explained in further detail below with respect to FIGS. **33** and **34**, the primer pocket swaging operation can include a case support die **108** in the fourth station **S4** configured to contact the case from above to brace the case against the force of the primer pocket swaging tool. When the resizing die (second station **S2**), the case support die **108** (fourth station **S4**), and tool head depth stop **104** are each properly adjusted, the tool head **18** has three points of contact with the case carrier and frame, which facilitates maintaining the tool head in a level orientation when the tool head is lowered. It will be appreciated that a similar effect is achieved even if there is contact at only two locations on generally opposing sides of the ram **46** (e.g., resizing and tool head depth stop).

[0095] Referring to FIG. **26**, the tool head depth stop **104** includes a depth stop body **104A** in the form of a threaded sleeve in a threaded opening in the tool head **18**. A bushing **104B** (e.g., made of brass) is received in the depth stop body **104A** to act as a bearing for movement of the guide **100** in the depth stop. The depth stop **104** is adjustable by changing the amount the depth stop body **104A** is threaded upward into the tool head **18**. The depth stop body **104A** includes a stop surface at its lower end configured to engage the frame **14** when the tool head **18** is lowered. The return spring **102** is nested in the lower end of the depth stop body **104A** such that the stop surface is outboard of the spring and the depth stop body transmits the spring force to the tool head **18**. The depth stop **104** includes a lock nut **104C** threaded on the depth stop body **104A** that can be tightened against the tool head **18** to secure the depth stop body in a desired position. Desirably, the location of the stop surface is adjusted by the operator to contact the frame **14** when the resizing die contacts the case carrier to counteract the moment induced by the resizing operation.

[0096] The tool head depth stop **104** can have other configurations or be omitted without departing from the scope of the present disclosure. For example, the tool head depth stop can be provided in a die mount of the tool head (e.g., at the seventh station **S7**) instead of at the tool head guide receiver. It will be appreciated that locating the tool head depth stop at the guide receiver frees the seventh station for incorporating a die to increase the available operations on the cases.

[0097] As shown schematically in FIG. **30**, the press system can include a control system **110** including a press controller **112** (e.g., microprocessor or central processing unit), a non-transitory tangible storage medium **114** (e.g., including forms of storage such as software and/or firmware),

and a user interface **116** (e.g., including a user input **116A** and an alarm **116B**). A power source **118** such as batteries or a port for a power cord (broadly, “electrical power input port”) can be used for providing electrical power to the control system **110**. The control system can include one or more LEDs **120** (broadly, “illuminators”) for illuminating work areas of the press (e.g., the ten stations **S1-S10**). The control system **110** also includes a plurality of accessory ports **122** to which press accessories can be connected. For example, in FIG. **30**, connected accessories include a low primer sensor **130**, a primer holder illuminator **132**, and a primer detector **134**, all described in further detail below. The ports **122** can include electrical power terminals and communication terminals for forming a power and communication connection with an accessory via a single port connection (e.g., micro USB). Alternatively, the power and communications can be provided separately (e.g., wireless communications, such as Bluetooth, Zigbee, etc., between accessory and press controller). The control system **110** also includes a communications port **140** for transmitting communication signals from the control system **110** to other devices and/or receiving communication signals. For example, the communications port **140** can be a wired port (e.g., USB or micro USB port, collectively “USB type” port) or a wireless port (e.g., WiFi, Bluetooth, Zigbee, etc.).

[0098] The control system **110** includes interconnection electronics **142** (e.g., including electrical, fiber optic lines, and/or wireless communication devices) that operatively connect the various components of the control system with each other and with other components of the press system **10**. It will be appreciated that the interconnection electronics **142** can include other components, such as A/D converters and/or filters through which signals such as the scale signal passes to the press controller.

[0099] Referring to FIGS. **27-29**, in the illustrated embodiment, the press system **10** includes a printed circuit board assembly **144** forming a part of the press control system **110**. The printed circuit board assembly **144** includes a printed circuit board structure **144A** (shown as a single circuit board, but optionally comprising multiple circuit boards directly or indirectly connected (e.g., tethered)), the press controller **112**, and the storage medium **114**. The printed circuit board includes interconnection electronics **142** in the form of circuitry operatively connecting the press controller **112**, storage medium **114**, and other components connected to the circuit board, such as the LEDs **120** and the accessory ports **122**. In this embodiment, the power source port **118** and communications port **140** are combined into a single port (e.g., micro USB port) having both power and communications terminals, and the single port is accessible at a side of the tool head **18** and can be used for connecting the onboard press controller **112** to an offboard controller or control system. The printed circuit board assembly **144** is receivable in a recess in the bottom of the tool head **18** that is sized and shaped to correspond to a shape of the printed circuit board structure **144A**. A plurality of fasteners with corresponding spacers (e.g., resiliently compressible rubber washers) are used to secure the printed circuit board assembly **144** in position and to secure a cover **146** (e.g., transparent or translucent plastic) over the printed circuit board assembly. The LEDs **120** face downward and emit light that passes through the cover to illuminate the stations **S1-S10**.

[0100] The accessory ports **122** (e.g., 3.5 mm sockets) are located on the printed circuit board structure **144A** to correspond to locations of openings **150** in the tool head **18**. When the printed circuit board assembly **144** is secured in the recess in the bottom of the tool head **18**, the accessory ports **122** are located to be accessible through the openings **150** in the top of the tool head. Suitable cords (e.g., cables having 3.5 mm jacks) can be used for connecting the accessories to the ports **122**. Plugs **152** can be used to close and protect the ports when the ports are not in use. Nine ports **122** are provided, but other numbers can be used without departing from the scope of the present disclosure.

[0101] The press controller **112** is configured to read and execute instructions stored in the storage medium **114**, and is responsive to the user interface **116**, for controlling operation of the press system. The press controller **112** receives signals from the accessory ports **122** (if used) and, based on instructions stored in the storage medium **114**, transmits responsive signals (e.g., command

signals) to one or more components of the press system **10**. In the illustrated embodiment, the user input **116A** comprises a push button actuator **116A'** for turning on/off the press controller **112** and the LEDs **120**. A user can enter and/or modify instructions stored on the storage medium **114** via the communications port **140**. The user interface **116** can include a display (not shown), and the LEDs **120** can be used as part of the user interface, as will become apparent. For example, when the press controller **112** determines a fault, it can energize the alarm **116B** (e.g., emit sound from a speaker) and/or flash and/or change colors (broadly, "change state") of the LEDs **120** to notify the operator of the fault. Example faults or malfunctions include case misfeed, primer misfeed, primer/bullet/case/powder refill needed, etc. Other types of user interfaces (e.g., including other types of user inputs) can be used without departing from the present disclosure.

[0102] In some embodiments, one or more press accessories connected to the ports **122** will include their own accessory controllers, such that the accessories can determine a fault condition based on sensory feedback and communicate that fault condition to the press controller **112**. Alternatively, the press accessory may merely transmit sensor signals to the press controller **112**, which determines a fault condition based on instructions in the storage medium **114**.

[0103] It will be appreciated that in other embodiments the press controller **112**, the storage medium **114**, and/or the user interface **116** can be part of another device such as a smart phone or tablet operatively connectable to the press system (e.g., wirelessly, such as via a wireless communications port of the control system) without departing from the scope of the present disclosure. Moreover, other configurations of press control systems can be used without departing from the scope of the present disclosure.

[0104] In view of the above, it will be appreciated that the printed circuit board assembly **144** serves as a harness for consolidating power and communications connections at the accessory ports **122**. Power to those ports **122** is provided via one connection to the upper tool head **18** at the micro USB port **118**, **140**. Moreover, if desired to transmit signals from the accessories and/or onboard controller **112** to an offboard control system, that can be done via the single micro USB port **118**, **140**. This limits the number of power supplies needed (e.g., a single USB to micro USB power cord) for powering various press accessories, and limits the overall number of cords needed, which is beneficial for avoiding interference with the moving components of the press system. The accessories mounted on the upper tool head **18** will move with their power and/or communication connections at the ports **122** and thus not tend to extend or pull on the cords connecting them for power and/or communications as the upper tool head moves. In other words, because the power and/or communications ports **122** are carried by the upper tool head **18**, the upper tool head in its up/down strokes does not move relative to these ports and is less likely to lead to inadvertent disconnection or obstruction. The printed circuit board assembly **144** serves as a power and communications hub carried by the press **12**, and in particular carried by the upper tool head **18**.

[0105] In another aspect of the present disclosure, the press **12** is configured to facilitate decapping operations (spent primer removal) at the second and third stations **S2**, **S3**. Redundant decapping operations can be implemented in sequential stations to more reliably remove spent primers from the cases. Two stations prior to the swaging/priming stations both allow primers to be ejected and drop into the spent primer collection receptacle.

[0106] A problem for conventional progressive ammunition presses is primer "draw-back." This occurs when the spent primer sticks to the end of the decapping pin (the pin that punches the spent primer out of the case) and is drawn back into the primer pocket as the decapping pin is extracted. The case is then moved to the next station, and the reloading machine attempts to either swage the primer pocket or seat a new primer on top of the spent primer, depending on the configuration of the machine. Draw-back is especially common when loading military cases with crimped primers, or other rounds that require excessive force to push the primer out of the primer pocket.

[0107] Current solutions to this issue include spring loaded decapping pins and vacuum tubes intended to use vacuum force to pull the spent primer off of the decapping pin before it can be

drawn back into the pocket. The vacuum systems require a vacuum source and are noisy. Spring loaded pins are delicate and break often. Presses with fewer stations usually require a single decapping operation (e.g., single decapping and resizing operation) to permit a single-pass process. [0108] In one example, a universal decapping die can be used in the second station S2, and a decapping and resizing die can be used in the third station S3. This provides a second chance to eject a stuck spent primer from a case (increasing reliability) and reduces chances of damaging the decapping and resizing die due to a particularly stubborn primer. Because there are ample stations on the press **12**, there is room to provide redundant decapping operations even in a single-pass processing operation. A second decapping pin will reliably separate the primer from the case using readily available and affordable standard decapping dies. The first decapping pin will overcome all of the force required to separate the primer from the primer pocket, and when the second pin contacts the spent primer it should fall away freely. The use of a redundant decapping die can also prevent damage to expensive resizing dies. If a universal decapping die is used in the first station and a combination sizing/decapping die is used in the second station, the universal decapping die will encounter any damaging obstructions first and will protect the more expensive and caliber specific resizing die.

[0109] Referring to FIG. **31**, the press **12** is configured to collect spent primers from both the second and third stations S2, S3. The frame **14** includes a cavity **120** sized and shaped to receive spent primers that may fall from the second and third stations S2, S3. Interior walls **122** of the frame form a funnel for funneling these spent primers to an outlet **120A** of the cavity. The press includes a tube connector **124** in the form of a tube receiver in which an upper end of a spent primer transport tube **126** can be installed (e.g., retained by friction). The arrangement is such that spent primers collected by the cavity **120** are funneled into the spent primer transport tube **126**. As shown in FIGS. **31** and **32**, the frame **14** includes an opening **14'** in the bottom of the frame and an opening **14''** in the side of the frame. The operator can choose to route the spent primer transport tube **126** through the bottom opening **14'** if desired to run the tube through the support surface (e.g., table top), or can choose to route the tube through the front opening **14''**. In either case the outlet end of the tube **126** can be located to deliver the spent primers to a spent primer receiver such as a bucket or other receptacle.

[0110] Referring to FIGS. **33** and **34**, the case support die assembly **108** can be installed at the fourth station S4 (swage) to support the case from above to oppose the force of the primer pocket swage tool as the primer pocket is swaged. The case support die assembly **108** includes a die body **108A**, which can be adjusted so that the bottom of the die is in contact with the case carrier **58** at the bottom of travel of the tool head **18**. The die body **108A** can be locked in place using a lock nut **108B** (e.g., 7/8-14 nut). The die assembly **108** includes a stem **108C** that can be adjusted (threaded into the die body **108A**) so that the stem contacts the bottom of the inside of the case at the bottom of travel of the tool head **18**, holding the case down against the swage pin. The position of the stem **108C** can be secured with respect to the die body **108A** using a stem lock nut **108D**.

[0111] As shown in FIGS. **35** and **36**, the lower tool head **20** includes a tool head body **20A** and two linear bearings **20B**, which act as guides movable along a track **130** formed by two guide rods secured to the frame. The lower tool head **20** is associated with the fourth and fifth stations S4, S5. The tool head body **20A** includes two threaded openings **20A'** (broadly, "tool mounts") for receiving tools to be associated with the fourth and fifth stations S4, S5. In the illustrated embodiment, a swaging pin **132** is mounted on the tool head body **20A** to be associated with the fourth station S4, and a priming pin **134** is mounted on the tool head body to be associated with the fifth station S5. The pins **132**, **134** can be adjusted by threading them into or out of the tool head body **20A**. Lock nuts **136** are provided to secure the pins **132**, **134** in their adjusted positions.

[0112] The drive system **22** includes a linkage connecting the right journal shaft to the lower tool head for moving it upward and downward. The linkage includes a link **140** having an opening with a bearing therein that is received over the cam **42A** of the right journal shaft **42**. An opposite end of

the link **140** is connected by a pivot connection to the lower tool head **20**. The cam **42A** is oriented such that the lower tool head **20** reaches the top of its travel when the ram **46** is at the bottom of its travel.

[0113] Referring to FIGS. **37-42**, the primer feeder **32** is mountable on the frame **14** to locate the primer feeder to feed primers for insertion to cases at the fifth station **S5**. The frame **14** includes a primer feeder mount **140** designed to receive the primer feeder **32** and permit the primer feeder to deliver primers to below the case carrier **58** for insertion into cases by the priming pin **134**. As shown in FIG. **37**, the primer feeder **32** is connected to the mount **140** by four fasteners. As will become apparent, the primer feeder **32** is designed to be self-contained, recirculating, and gear-driven.

[0114] The primer feeder **32** includes a feeder body **142** that serves as a support for other components of the primer feeder. The feeder body **142** includes a mount **142A** for holding a primer supply tube assembly **144**. The feeder body **142** also serves as a support for a primer gear **146** (broadly “primer shuttle” or “primer wheel”) configured to receive primers from the primer supply tube assembly **144** and deliver them to the priming pin **134**.

[0115] The primer supply tube assembly **144** is shown in closer detail in FIGS. **38** and **39**. The tube assembly **144** includes an inner primer feed tube **144A** made of plastic or another suitable material. The tube assembly **144** also includes an outer shield tube **144B** (broadly, “shield”). Desirably, the outer shield tube **144B** is made of metal or another suitable material for resisting lateral explosion if primers in the tube assembly ignite. Such an explosion desirably exits the tube assembly upward rather than laterally to reduce potential for injury to the user. Referring to FIG. **39**, the tube assembly **144** includes a threaded cap **144C** that threads onto the shield tube **144B** and engages a top end of the primer feed tube **144A** to secure the primer feed tube in position. As shown in FIG. **41**, the bottom end of the primer feed tube **144A** has a reduced diameter creating a shoulder that rests on a lip of the feeder body **142**. The lower end of the shield tube **144B** is threaded to the feeder body **142** to secure the tube assembly to the feeder body.

[0116] As shown in FIG. **41**, the primer gear **146** has a central opening receiving a bearing **150**. The primer gear **146** is secured to the feeder body **142** by a fastener and an associated washer. The primer gear **146** rotates about the longitudinal axis of this fastener. The primer gear **146** has four primer holders **152** (e.g., stepped recesses having shoulders for supporting primers therein) spaced around the primer gear at intervals of about 90 degrees. The primer gear **146** is rotatable about the fastener to present a primer holder **152** to receive a primer from the bottom of the primer tube assembly **144**. As shown in FIG. **40**, a ball detent **154** and associated spring are captured in a pocket in the feeder body **142** and configured to nest in index openings **156** spaced at 90 degree intervals around the primer gear **146**. The ball detent **154** facilitates locating the primer holders **152** in proper position to receive primers from the primer tube assembly **144** and for presenting primers to be inserted in cases.

[0117] The primer gear **146** has teeth **146A** configured to mesh with the teeth **62A** of the case carriage gear **62** such that the primer gear is driven by the gear of the case carriage. Accordingly, each time the case carrier **58** is indexed, the primer gear **146** is indexed too. The gears **62**, **146** are sized so that as the case carrier **58** is indexed forward one station, the priming gear rotates 90 degrees, which causes a new primer to be located directly between the priming pin **134** and the center axis of the case in the fifth station **S5**. When the ram **46** is lowered, the priming pin **134** presses the new primer into the bottom of the case.

[0118] The primer feeder **32** includes a clutch spring **160** installed over a boss of the priming gear **146**. The clutch spring **160** has a looped end **160A** secured to the feeder body **142** by a fastener. A second end **160B** of the clutch spring **160** is captured by a lever **162** that has an opening received over the boss of the priming gear **146**. A coil **160C** of the clutch spring **160** wraps around the boss of the primer gear **146**. The arrangement is such that the coil **160C** permits clockwise rotation of the priming gear **146**. However, counter-clockwise rotation of the priming gear **146** is prevented

because such motion of the priming gear boss in the clutch spring coil **160C** causes the coil to constrict the boss. It will be appreciated that the case carrier **58** is likewise prevented from turning backwards (clockwise direction for the case carrier) because its gear **62** is in mesh with the primer gear **146** and is effectively stopped by the clutch spring **160**. If desired, the user can release the clutch spring **160** by pulling the lever **162** laterally, which causes the lever to pivot about the primer gear boss and to bring the associated end **160B** of the clutch spring with it, thus loosening the coil **160C** on the hub. The primer gear **146** and case carrier **58** can be turned backward while the user holds the release lever **162** against the bias of the clutch spring **160**.

[0119] In another aspect of the clutch spring **160**, the spring acts as an index damper for the case carrier **58**. Providing damping to the case carrier movement can help prevent powder spillage and reversal of case carrier **58**. As explained above, the case holders **64** of the case carrier are located in the stations by engagement of a ball detent **72** with notches **58A** in the case carrier. When the detent ball **72** begins to engage a notch **58A** in the case carrier **58**, the case carrier may jump forward abruptly and outrun the index mechanism. This can cause the case carrier **58** to jerk the cases and result in spilled powder, which reduces the consistency of the powder charge weights and can clog the press and affect the indexing and priming functions. Rotating the case carrier backwards (while sometimes necessary to correct a malfunction) can lead to dangerous conditions such as double charges of powder and crushed primers. The clutch spring **160** on the primer gear **146** induces a small amount of drag on the case carrier **58**. This prevents the case carrier from “jumping” ahead of the index mechanism and reduces powder spillage. The friction of the clutch spring **160** on the boss of the primer gear **146** is transmitted to the case carrier **58** via the meshed teeth of the gears. It will be appreciated that the clutch spring **160** can be referred to as a case carrier movement damper or rotation damper.

[0120] In one aspect of the primer gear **146**, the primer holders **152** are separated from each other sufficiently to prevent a chain reaction in the event that a primer is inadvertently ignited in the primer gear. For example, if a primer in position to be pushed into a case is inadvertently ignited (e.g., by the priming pin), that primer is separated from the other primers in the primer gear, and from all of the primers in the primer tube assembly. The ignited primer would not generate a chain reaction of primers igniting up the primer tube assembly **144**.

[0121] It will be appreciated that the primer feeder **32** can be removed from the frame **14** as a self-contained unit. The primers do not need to be removed from the primer tube assembly **144** before removing the primer feeder **32**. When the primer feeder **32** is disconnected, the ball detent **154** of the primer feeder engaging the primer gear **146** holds the primer gear in position to prevent the gear from inadvertently rotating and dispensing primers. Desirably, a kit of two primer feeders **32** is provided with the press **12** such that the operator can change between a first primer feeder for feeding small primers and a second primer feeder for feeding large primers. The large primer feeder would have essentially the same construction as shown in FIGS. **37-41** but would have an inner feed tube **144A** of larger diameter, and the primer holders **152** of the primer gear would be larger. Accordingly, the process of changing the press system over from loading different types of cases can be made simpler by permitting the user to change out the self-contained primer feeder **32**. This is more convenient than conventional systems in which converting the priming system between large and small primers is difficult and time consuming.

[0122] As mentioned above, each advancement of the case carrier **58** causes the primer gear **146** to advance a quarter turn. Thus, the primer gear **146** moves the four primer holders **152** through four stations **P1-P4**. In the first station **P1**, the primer holder **152** is located under the primer tube assembly **144** for receiving a primer. In the third station **P3**, the primer is located under the case carrier **58** and above the priming pin **134** for installing the primer in a primer pocket of a case. The second and fourth primer stations **P2, P4** serve as valuable witness stations. As shown in FIG. **42**, the feeder body **142** includes first and second witness openings **166** corresponding to the second and fourth primer stations **P1, P4**. The witness openings **166** permit the user to see the primer

holders **152** in the second and fourth primer stations **P2**, **P4**. The second primer station **P2** serves as a witness station where the operator can visually confirm that a primer is present in the primer holder **152** (i.e., that a primer has successfully loaded into the primer holder from the feed tube). Thus, the user can confirm a primer is “on deck” to be installed in a future case. The fourth primer station **P4** serves as a witness station for the operator to visually confirm that a primer is not present in the primer holder (i.e., that the primer was successfully loaded into the case). If primer insertion did not happen properly, the primer may have been crushed and remains in the primer holder **152** at the fourth station **P4**. The fourth station provides the operator an opportunity to remove the damaged primer from the primer gear **146**. If a case was not present when the priming pin **134** pushed the primer upward, the primer would return downward to the primer holder **152** and would subsequently travel to the fourth primer station **P4**. An undamaged primer in the fourth priming station does not need to be removed. The primer would remain in the primer holder **152** as it turns under the primer tube assembly **144** and would prevent another primer from falling into the primer holder. The primer gear **146** can be referred to as “recirculating” because it recirculates primers not picked up by a case.

[0123] Referring to FIGS. **47** and **48**, the witnessing of the second and fourth primer stations **P2**, **P4** can be automated. As shown in FIG. **48**, a first circuit board assembly **170** can be configured to include a circuit board structure **172**, two photo sensors **174** (broadly, “primer detectors” **134**), a power and communication port **176** (e.g., 3.5 mm socket), and circuitry operatively connecting these components. A second circuit board assembly **180** can be configured to include a circuit board structure **182**, two LEDs **184** (broadly, “primer holder illuminators” **132**), a power and communication port **186** (e.g., 3.5 mm socket), and circuitry operatively connecting these components. Referring to FIG. **47**, the first circuit board assembly **170** is mounted by fasteners to the feeder body **142** such that the photo sensors **174** overlie and look downward toward the primer holders **152** in the second and fourth primer stations **P2**, **P4**. The second circuit board assembly **180** is mounted by fasteners to the frame **14** such that the LEDs **184** face upward and are in registration with openings in the frame below the second and fourth primer stations **P2**, **P4**. The light from the LED **184** associated with the second primer station **P2** should always be blocked by a staged primer, and the light from the LED **184** in the fourth primer station **P4** should always be able to pass through the primer holder **152** to be sensed by the photo sensor **174**. Any failure to satisfy one of these conditions would result in an alarm (or a stoppage if the automation kit is attached). The tangible storage medium **114** of the press control system **110** can include instructions to signal an alarm, such as by strobing the LEDs **120**, changing the colors of the LEDs **120**, and/or energizing an audible alarm **116B**, when one of the conditions is not met. The photo sensors **174** detect the presence or absence of a primer by the absence or presence of light.

[0124] Referring to FIGS. **30** and **48**, the first and second circuit board assemblies **170**, **172** can be connected to respective ports **122** of the upper tool head **18** for providing power thereto and communication of the sensor signal to the press controller **112**. Alternatively, the first and second circuit board assemblies **170**, **180** could be tethered together with one of them (e.g., the first circuit board assembly) being connected to a port **122** of the upper tool head **18**, for providing power to both of the circuit board assemblies, and communication from the photo sensors **174** to the press controller **112**.

[0125] It will be appreciated that other types of primer detectors can be used without departing from the scope of the present disclosure. For example, one or more mechanical switches can be used instead of an optical switch or photo sensor.

[0126] Referring to FIGS. **43-46**, the primer feeder **32** can be configured to automatically indicate whether the stack of primers remaining in the primer feed tube assembly **144** is low or has run out. This can be accomplished by installing a primer follower **190** in the primer feed tube **144A**. The follower **190** has a bottom end that rests on the top of the stack of primers. The primer follower has an upper end that includes a cylindrical magnet **190A**. In the illustrated embodiment, the magnet

190A is overmolded to capture the magnet and connect it to an upper end of a shaft of the follower. As primers are removed from the bottom of the stack, the stack of primers shortens, and the primer follower moves downward.

[0127] A low primer sensor **130** is provided on the upper end of the primer feed tube assembly **144** for sensing when the magnet **190A** moves near the low primer sensor. In the illustrated embodiment the low primer sensor **130** includes a sensor support body **192** having first and second portions secured together by fasteners. The body portions capture a printed circuit board assembly **194** which includes a printed circuit board structure **194A**, a Hall effect sensor or magnetic Hall switch **194B**, a power and communication port **194C** (e.g., 3.5 mm socket), and circuitry operatively connecting the components. When the body portions are secured to each other, the body **192** captures the upper end of the primer tube assembly, and the port **194C** is accessible to permit connection of a cord **196** (e.g., cable having opposite ends including 3.5 mm jacks). As shown in FIG. **30**, the opposite end of the cord **196** can be connected to a port **122** of the upper tool head for providing power to the printed circuit board assembly **194** and providing communication to the press controller **112**.

[0128] When the stack of primers reduces in height, eventually the magnet **190A** of the primer follower **190** becomes close enough to the Hall effect switch **194B** to cause it to change state, which is communicated to the press controller via the cord **196**. The press controller **112** executes instructions stored on the tangible storage medium **114** to, after the change in state is detected, generate an alarm, such as by strobing the LEDs **120**, changing the color of the LEDs **120**, and/or energizing an audio alarm **116B**. When the user removes the primer follower **190** to install more primers, the magnet **190A** moves away from the sensor **194B** and the alarm stops, and the cycle begins again. It will be appreciated that the low primer sensor **130** does not have a mechanical fatigue limit and should yield long life. Optionally, the sensor signal could be communicated to a control system component offboard the press via the communications port.

[0129] Referring to FIGS. **49-52**, the powder dispenser **34** will be described in further detail. The powder dispenser includes a threaded stem **200** and a powder hopper **202** supported by the stem. The powder hopper **202** is configured to funnel powder into a powder measure **206**. The powder measure includes a drum **208** and a micrometer **210**. The micrometer **210** is adjustable to change a position of a plunger **210A** in the drum **208** to change a volume of a cavity in the drum for receiving powder. The hopper **202** and powder measure **206** are connected to a powder delivery tube **214**. The powder delivery tube **214** has a shoulder **214A** that rests on an upper end of the stem **200** to locate the tube with respect to the stem. A lower end of the powder delivery tube **214** extends out of the bottom end of the stem **200** and is positioned to be engaged by a case mouth. For example, the stem **200** could be installed in the upper tool head **18** in the die mount **38** corresponding to the seventh station **S7**. When the upper tool head **18** is lowered, the lower end of the powder delivery tube **214** (broadly, “activator rod”) engages the case mouth, which causes the powder delivery tube to be pushed upward. This raises the shoulder **214A** of the powder delivery tube off the upper end of the stem **200** and raises the drum **208** and hopper **202**. This movement causes links **220** connected off-center to the drum **208** and to the stem to cause the drum to rotate to raise the powder measure **206**, shown by comparison of FIGS. **49** and **50**. Referring to FIG. **51**, when the powder measure **206** is pivoted upward, a mouth of the drum **208** opens to a chute below the drum leading to the powder delivery tube **214** for delivery of the powder to the case below.

[0130] A guide rod **216** connected to the hopper **202** by two fasteners extends through an opening of the stem **200**. A compression spring **218** on the guide rod **216** is captured between the stem **200** and a spring retainer **220** secured to the guide rod. The arrangement is such that when the hopper **202** is raised to cause the powder measure **206** to dispense powder, the guide rod **216** moves upward and the spring retainer **220** compresses the spring **218**. The compression of the spring **218** and downward bias on the spring retainer **220** facilitates return of the powder measure **206** to the lowered position when the upper tool head **18** is raised. In some instances, the spring **218** may not

provide sufficient force to reset the powder measure **206**. For example, powder may block the drum **208** from pivoting. The powder dispenser **34** of the includes a reset feature to ensure the powder measure **206** resets even if the spring is not able to reset the powder measure. Referring to FIG. **52**, a lower end of the guide rod **216** passes through an opening in the frame **16**. Two nuts **224** are threaded onto the guide rod **216** below the opening in the frame. The nuts **224** are adjusted such that the upper nut is positioned (and the lower nut locks the position) to crash into the frame **16** if the powder measure **206** has not reset and the upper tool head **18** is moving upward. The engagement of the frame **16** with the nut **224** holds the guide rod **216** in position as the powder measure **206** moves upward, thus forcing the drum **208** to rotate and reset the powder measure. This will prevent the press from loading subsequent rounds with the powder measure **206** stuck in the activated position. Rounds loaded while powder measure is stuck will not receive powder, and can cause “squibs,” which are rounds that do not have enough gas pressure to propel the bullet completely through the barrel. Squib rounds can be extremely hazardous to the shooter if they are not immediately recognized and subsequent rounds are fired.

[0131] Existing powder measures use a simple cone shape to funnel the powder through the outlet and into the case. This type of funnel can cause issues such as “bridging,” “rat-holing,” and “flooding” when used with coarse powders. The funnel **230** of the present disclosure a parabolic shape and does not have sharp edges that could cause stagnant flow and lead to such issues. The parabolic funnel **230** allows coarse powders to feed into cases as small as .20 caliber.

[0132] It will be appreciated that powder dispensers having other configurations can be used without departing from the scope of the present disclosure. Moreover, aspects of the powder dispenser can be used in benchtop powder measures (not on a press), and a stand-alone powder funnel kit, without departing from the scope of the present disclosure.

[0133] With reference to FIGS. **53-54**, the press **12**, and in particular the upper tool head **18**, includes a provision for an integrated bullet feeder **36**. The tool head **18** has clearance and mounting provisions for a proprietary bullet feeding system **36** which enables feeding and completely seating bullets in the same station **S9**.

[0134] In conventional case activated bullet feeders, the feeder threads into one of the die stations on the tool head, and places a bullet on top of a case as the tool head is lowered. The bullet is then required to ride on top of the case to the next station as the machine indexes, and is seated into the case by the next stroke of the ram. Requiring the bullet to ride on top of the case between stations creates the potential for the bullet to tip and crush the case as it is seated, or to fall off completely, potentially resulting in powder spillage and waste.

[0135] The bullet feeder **36** of the present disclosure attaches to the bottom of the upper tool head **18**, and uses a shuttle **254** to place the bullet directly under the seating stem **241** of the seating die prior to the tool head **18** being lowered. As shown in FIG. **4**, the upper tool head **18** includes a bullet feeder mount **242** including three openings for receiving fasteners to secure a bullet feeding tube **244** to the top of the tool head. The tool head **18** also includes a bullet feeder opening **246** to allow passage of bullets from the tube **244** to the shuttle below the tool head. The bottom of the tool head **18** includes openings for receiving fasteners to secure a shuttle assembly **250** of the bullet feeder **36** to the bottom of the tool head **18**. The shuttle assembly **250** includes a shuttle mount **252** and a shuttle **254** movable with respect to the shuttle mount. The shuttle mount **252** includes a track **256** and a compression spring **258** biasing the shuttle toward a bullet receiving position in registration with the bullet feeder opening **246**. As the upper tool head **18** moves downward, a bearing **260** (broadly, “wheel”) carried by the shuttle **254** engages a ramp **262** that causes the shuttle to move to position the bullet in registration with the bullet seating stem **202** in the ninth station **S9**. The shuttle **254** includes a spring biased pin **264** (broadly, “bullet retainer”) that functions to hold the bullet in the shuttle but permits the case to deflect the pin so the case can move upwardly past the pin to push the bullet against the seating stem **241** to seat the bullet into the case mouth. When the tool head **18** is raised, the bullet is held by the case and withdrawn from the

shuttle **254** past the pin **224**, and the shuttle returns to receive another bullet and the cycle repeats. Accordingly, bullet feeding and seating occurs at the same station **S9**.

[0136] The shuttle mount **252** can include an adjustable limiter **270** to accommodate different size bullets so that the shuttle **254** can move laterally without pinching the top of the bullet in the shuttle or the bottom of the next bullet in the feeder tube. The height of the limiter **270** can be adjusted by adding or removing spacers **272** between the limiter and the tool head **18**. As the shuttle **254** moves under the seating station **S9**, the bullet drops off of the limiter **270** and onto the detent pin **264**.

[0137] If the press **12** is cycled with no case in the bullet feeding station **S9**, the shuttle **254** will return to the feeder tube station with the bullet still sitting on the detent pin **264**. The shuttle **254** will be pushed rearward until the bullet contacts the limiter **270**. This will prevent the shuttle **254** from picking up an additional bullet from the feeder tube, but the spring **258** will not exert enough force to damage the bullet in the shuttle.

[0138] It will be appreciated that other configurations of bullet feeders can be used without departing from the scope of the present disclosure. For example, the limiter **270** can be omitted. Moreover, the shuttle may be located above the tool head **18** and deliver a bullet into a seating die (e.g., a side window in the seating die) from above the tool head to fall into position under the seating stem.

[0139] An alternative embodiment of a bullet feeder **280** is shown in FIGS. **55-58**. In this embodiment, a bullet feeding tube **282** is connected to the top of the tool head **18** as in the previous embodiment. Likewise, this embodiment includes a shuttle assembly **284** including a shuttle mount and a shuttle **288**. The shuttle mount is not shown in FIGS. **55-58**, but it will be understood the shuttle mount is similar to the mount **252** described above, including a track and associated spring. The shuttle **288** is moveable from registration with the bullet feeding tube **280** to registration with the bullet seating stem **241** at the ninth station **S9** in the same manner described above (bearing engaging ramp). In this embodiment, the shuttle **288** has a different configuration. The shuttle **288** includes a shuttle body **290** and a bullet holder **292** carried by the shuttle body. The bullet holder **292** includes a bullet cradle **294** and bullet gates **296** connected to the bullet cradle. The bullet cradle **294** is connected to the shuttle body **290** by two fasteners. Springs **300** on the fasteners are captured between heads of the fasteners and the shuttle body **290** such that the bullet holder **292** is biased downward into engagement with shoulders of the shuttle body, but the bullet holder is moveable upward against the bias of the springs. The bullet gates **296** are pivotable with respect to the bullet cradle **294** about pivot connections **302**. Torsion springs **304** are provided on the pivot connections to bias the gates **296** toward bullet retaining positions. Pins **306** connected to the bullet cradle **294** are received in slots of the bullet gates **296** to limit movement of the gates under the bias of the torsion springs **304**.

[0140] When a bullet falls into the bullet cradle **294**, the bullet rests on the bullet gates **296**, as shown in FIG. **56**. When the shuttle **288** moves to position the bullet in registration with the seating stem **241** at the ninth station **S9**, the next bullet in the stack falls onto an upper surface of the shuttle **288**, as shown in FIG. **56**. Referring to FIG. **57**, when the tool head **18** moves downward, the case engages the gates **296** and deflects them laterally out of the way. The case carries the bullet upward for engagement with the seating stem **241**. Desirably, the cradle **294** has a relatively tight fit to the bullet to prevent the bullet from tipping. As shown in FIG. **58**, as the bullet is pressed into the mouth of the case to the proper seating depth (adjustable via adjustment of the seating die), the bullet holder **292** moves upward. The case mouth engages the bottom of the bullet cradle **294** and causes the bullet holder **292** to move upward against the bias of the springs **300**. When the upper tool head **18** is raised, the bullet is held by the case and withdrawn from the gates **296**, and the gates return to the bullet retaining positions. When the shuttle **288** returns to locate the bullet holder **292** in registration with the bullet feeder tube **280**, the next bullet falls from the top of the shuttle into the bullet cradle **294** onto the gates **296**, and the cycle repeats.

[0141] Referring to FIG. **59**, the press **12** includes a plurality of covers. The covers can be

connected to sides of the frame **14** by fasteners to cover openings in the frame. In the illustrated embodiment, three one-piece covers **330** are installed on the front, left, and rear of the frame. A two-piece cover is installed on the right side of the frame. The top section **332A** of the right hand cover can be removed to allow adjustment of the swaging and priming pins without removing the lower section **332B** of the cover. The covers **330**, **332A**, **332B** can comprise translucent or transparent material (e.g., plastic) to permit the operator to observe the function of the press components but be protected from them.

[0142] As shown in FIG. **60**, the press system includes a bracket **340** connectable to the frame **14** by fasteners. The bracket **340** is configured to carry a bin **344** and to locate the bin to receive loaded cartridges ejected from the case carrier **58**.

[0143] In one aspect of the press system **10**, the drive system **22** is configured to receive back-and-forth drive input, or receive 360 degree rotation input, with no change in operation of the press **12**. It will be appreciated that if the stop screws **56** (FIG. **6**) are removed from the frame **14**, the crank throws **50** become unobstructed and are permitted to rotate 360 degrees rather than 180 degrees. Accordingly, the user can rotate the lever **26** in 360 degrees instead of in a back-and-forth or down-and-up fashion. A full 360 degrees of rotation causes the drive system **22** to output movement to the press system components in the same manner as back-and-forth operation described above. As the lever **24** turns in the first 180 degrees of motion, the tool heads **18**, **20** move toward the case carrier **58**, and as the lever turns in the next 180 degrees of motion, the tool heads move away from the case carrier. Most likely, if the press **12** is to be used in a manual mode of operation, the 180 degrees range of motion will be desirable. But if the press is to be operated by a motor, the 360 degrees of motion is desirable.

[0144] An example of an automatic press actuation system **26** is shown in FIG. **62**. The actuation system **26** includes a support **360** to which a motor **28** is connected. A first wheel **364** is mounted on an output shaft of the motor **362**. The first wheel **364** is operatively connected to a second wheel **368** that is connected to the hub of the right journal shaft **42**. Although the wheels **364**, **368** are shown as geared pulleys connected via a belt **370**, it will be appreciated the wheels could be smooth pulleys used with a belt, could be sprockets used with a chain, or could be gears in mesh with each other (without belt or chain). Other configurations could be used without departing from the scope of the present disclosure.

[0145] With an automatic press actuation system **26**, the main drive input can be a continuous 360° motion. This offers the following advantages for automation. The control software can be simplified because the controller does not reverse the input motion at each end of the stroke and therefore does not need to know the exact position of the crank to determine when to reverse the input. The machine runs smoother, because it does not have to load and unload all of the play in the system at the end of each stroke. The ram is guaranteed to achieve both top dead center and bottom dead center on each stroke.

[0146] In one example, the press system **10** could be set up in the following manner (e.g., to load .223 caliber ammunition). The first station **S1** could include a conventional case feeder. The second station **S2** could include a universal decapping die on the upper tool head **18**. The third station **S3** could include a decapping and resizing die, or just a decapping die, on the upper tool head **18**. The fourth station **S4** could include a swaging tool **132** on the lower tool head **20** and the case support die **108** on the upper tool head **18**. The fifth station **S5** could include the gear driven primer feeder **32** discussed above on the upper tool head **18** and a priming pin **134** on the lower tool head **20**. The sixth station **S6** could include a trimmer on the upper tool head **18**. The seventh station **S7** could include the powder dispenser **34** on the upper tool head **18**. The eighth station **S8** could include a powder check or a blank die (e.g., depth stop die). The ninth station **S9** could include the bullet feed and seat operation described above. The tenth station **S10** could include a crimp die or secondary seating die (redundant).

[0147] Embodiments of the aspects of the invention may be described in the general context of data

and/or processor-executable instructions, such as program modules, stored on one or more tangible, non-transitory storage media and executed by one or more processors or other devices. Generally, program modules include, but are not limited to, routines, programs, objects, components, and data structures that perform particular tasks or implement particular abstract data types. Aspects of the invention may also be practiced in distributed computing environments where tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules may be located in both local and remote storage media including memory storage devices.

[0148] In operation, processors, computers and/or servers may execute the processor-executable instructions (e.g., software, firmware, and/or hardware) such as those illustrated herein to implement aspects of the invention.

[0149] Embodiments of the aspects of the invention may be implemented with processor-executable instructions. The processor-executable instructions may be organized into one or more processor-executable components or modules on a tangible processor readable storage medium. Aspects of the invention may be implemented with any number and organization of such components or modules. For example, aspects of the invention are not limited to the specific processor-executable instructions or the specific components or modules described herein. Other embodiments of the aspects of the invention may include different processor-executable instructions or components having more or less functionality than illustrated and described herein.

[0150] The order of execution or performance of the operations in embodiments of the aspects of the invention illustrated and described herein is not essential, unless otherwise specified. That is, the operations may be performed in any order, unless otherwise specified, and embodiments of the aspects of the invention may include additional or fewer operations than those disclosed herein. For example, it is contemplated that executing or performing a particular operation before, contemporaneously with, or after another operation is within the scope of aspects of the invention.

[0151] When introducing elements of aspects of the invention or the embodiments thereof, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements.

[0152] In view of the above, it will be seen that several advantages of the aspects of the invention are achieved and other advantageous results attained.

[0153] Not all of the depicted components illustrated or described may be required. In addition, some implementations and embodiments may include additional components. Variations in the arrangement and type of the components may be made without departing from the spirit or scope of the claims as set forth herein. Additional, different or fewer components may be provided and components may be combined. Alternatively or in addition, a component may be implemented by several components.

[0154] The above description illustrates the aspects of the invention by way of example and not by way of limitation. This description enables one skilled in the art to make and use the aspects of the invention, and describes several embodiments, adaptations, variations, alternatives and uses of the aspects of the invention, including what is presently believed to be the best mode of carrying out the aspects of the invention. Additionally, it is to be understood that the aspects of the invention are not limited in its application to the details of construction and the arrangement of components set forth in the description or illustrated in the drawings. The aspects of the invention are capable of other embodiments and of being practiced or carried out in various ways. Also, it will be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

[0155] It will be apparent that modifications and variations are possible without departing from the scope of aspects of the invention as defined in the appended claims. It is contemplated that various changes could be made in the above constructions, products, and methods without departing from

the scope of aspects of the invention. In the preceding specification, various embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the aspects of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

Claims

1-16. (canceled)

17. A progressive ammunition press system comprising: a frame configured to be supported by a support surface; a case carrier supported by the frame, the case carrier including multiple case holders each configured to hold an ammunition case, the case carrier being moveable with respect to the frame to move the case holders to a plurality of stations; a tool head supported by the frame, the tool head comprising a plurality of press accessory mounts, the tool head being moveable with respect to the case carrier; a drive system operatively connected to the tool head to move the tool head with respect to the case carrier; a communications output port supported by and moveable with the tool head; and a communications input port supported by and moveable with the tool head and operatively coupled to the communications output port to transmit communication signals to the communications output port.

18. The progressive ammunition press system of claim 17, wherein the tool head includes an opening corresponding to the communications input port and the communications input port is accessible via the opening.

19. The progressive ammunition press system of claim 17, wherein the opening is located in an upper surface of the tool head.

20. The progressive ammunition press system of claim 17, further comprising a printed circuit board structure, the communications output port and communications input port being connected to the printed circuit board structure, the printed circuit board structure including circuitry operatively coupling the communications input port to the communications output port.

21. The progressive ammunition press system of claim 20, further comprising at least one LED connected to the printed circuit board structure and configured to receive electrical power from the communications output port, the at least one LED configured to emit light downward to illuminate an area below the tool head.

22. The progressive ammunition press system of claim 20, wherein the printed circuit board structure is connected to a bottom of the tool head.

23. The progressive ammunition press system of claim 20, wherein the printed circuit board structure comprises a single printed circuit board.

24. The progressive ammunition press system of claim 17, further comprising a press controller and a tangible storage medium each supported by the tool head, the tangible storage medium storing press controller executable instructions to generate a command signal to control an operation of the press system based on a sensor signal received from the communications input port.

25. The progressive ammunition press system of claim 24, wherein the press system includes an audible alarm, and the operation of the press system comprises energizing the audible alarm.

26. The progressive ammunition press system of claim 24, wherein the press system includes at least one illuminator, and the operation of the press system comprises changing a state of the illuminator.

27. The progressive ammunition press system of claim 26, wherein the illuminator is oriented to illuminate the case carrier.

28. The progressive ammunition press system of claim 17, wherein the communications signal input port includes an electrical power terminal.

- 29.** The progressive ammunition press system of claim 17, wherein the communications input port comprises a 3.5 mm socket.
- 30.** The progressive ammunition press system of claim 17, wherein the communications output port comprises a USB type connector.
- 31.** The progressive ammunition press system of claim 17, wherein the communications input port is a first communications input port, and the progressive ammunition press system further comprises a second communications input port supported by and moveable with the tool head and operatively coupled to the communications output port to transmit communication signals to the communications output port.
- 32.** The progressive ammunition press system of claim 17, wherein the communications input port is a first communications input port, and the progressive ammunition press system further comprises a second, third, and fourth communications input ports supported by and moveable with the tool head and operatively coupled to the communications output port to transmit communication signals to the communications output port.
- 33-37.** (canceled)
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